

CPIT-250 System Analysis
mid 22.5

Q1: MCQs

1. Is a data flow diagram that represents a system's major processes, data flows, and data stores at a high level of detail:
A. Level-0 diagram.
B. Context diagram.
C. DFD diagram.
2. Work or action performed on data:
A. Process.
B. Data store.
C. Source/link.
3. All functional features of the system chosen for development in analysis:
A. Logical design.
B. Physical design.
C. Design.
4. Is an iterative process of breaking a system description down into finer and finer detail:
A. Functional decomposition.
B. primitive DFD.
C. level-1 diagram.
5. If a DFD contains data flows that do not lead anywhere then the DFD is:
A. Completeness
B. Consistency
C. Timing
6. Data flow that appears on a higher level of DFD but not on lower level:
A. Violation consistency
B. Violation completeness
C. Iterative development
7. Refine understanding of system requirements in concrete terms:
A. System prototypes
B. CASE tools
C. JAD

Q2: Fill the blank

1. The primary purpose of using **JAD** in the analysis phase is to collect systems requirements simultaneously from the key people involved with the system.
2. **Observation** can cause people to change their normal operating behaviors.
3. When a JAD is completed, the end result is a set of **Documents** that detail the working of the current system related to the study of the faith replacement system.
4. The lowest level of a DFD is **Primitive**.
5. **Open-ended questions** are a question that have no prespecified answers.
6. **Data store** is a data at rest inside the system.
(التعريف حقو ما جا زي كذا بالضبط حسب ما أذكر انو مثلاً تمثيل الداتابيس في الرسمة تدخل في اي عنصر من الرسم؟ شي زي كذا)
7. **Planning** an organization total information system needs.
8. RUP established four phase of development: inception, elaboration, construction, and **transition**.

Q3: compare between.

(اذكر شيء واحد بس من الفروقات اللي بينهم)

	Agile Methods	Traditional Waterfall Methods
Size	Well suited for small teams and products.	Evolved to handle large teams and products.
Criticality	Untested for safety-critical products.	Specialized for highly critical products.
Dynamism	Benefits from simple design and continuous improvement.	Relies on detailed plans and upfront design.
Personnel	Requires a continuous presence of scarce experts.	Requires a critical mass of experts during definition.
Culture	Thrives in a culture that values freedom and flexibility.	Thrives in a culture that values clear roles and order.

Q4: mention two of

1. JAD participants:

- Managers.
- Users.

2. Disadvantaged of individual interviews:

- Contradictions and inconsistencies between interviewees.
- Follow-up discussions are time consuming.

Q5: Rules of DFD.

انقسمت قسمين

النوع الأول: ترسم القاعدة بحسب وصفها.

النوع الثاني: تصحح القاعدة لو اترسمت غلط وتذكر التعديل.

هذي اللي جات من القواعد : B - D - K - M

Data Flow Diagramming Rules (Cont.)

TABLE 7-2 Rules Governing Data Flow Diagramming

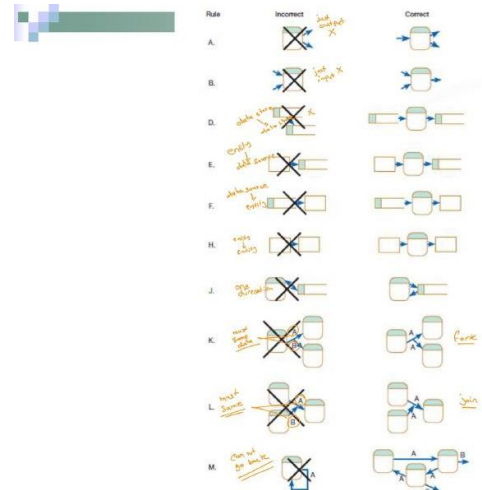
Process:
A. No process can have only outputs. It is making data from nothing (a miracle). If an object has only outputs, then it must be a source: 
B. No process can have only inputs (a black hole). If an object has only inputs, then it must be a sink: 
C. A process has a verb phrase label.
Data Store:
D. Data cannot move directly from one data store to another data store. Data must be moved by a process.
E. Data cannot move directly from an outside source to a data store. Data must be moved by a process that receives data from the source and places the data into the data store.
F. Data cannot move directly to an outside sink from a data store. Data must be moved by a process.
G. A data store has a noun phrase label.
Source/Sink:
H. Data cannot move directly from a source to a sink. It must be moved by a process if the data are of any concern to our system. Otherwise, the data flow is not shown on the DFD.
I. A source/sink has a noun phrase label.

Data Flow Diagramming Rules (Cont.)

TABLE 7-2 Rules Governing Data Flow Diagramming (cont.)

Data Flow:
J. A data flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The latter is usually indicated, however, by two separate arrows because these happen at different times.
K. A fork in a data flow means that exactly the same data goes from a common location to two or more different processes, data stores, or sources/sinks (this usually indicates different copies of the same data going to different locations).
L. A join in a data flow means that exactly the same data come from any of two or more different processes, data stores, or sources/sinks to a common location.
M. A data flow cannot go directly back to the same process it leaves. There must be at least one other process that handles the data flow, produces some other data flow, and returns the original data flow to the beginning process.
N. A data flow to a data store means update (delete or change). 
O. A data flow from a data store means retrieve or use. 
P. A data flow has a noun phrase label. More than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.

(Source: Adapted from celko, 1987.)

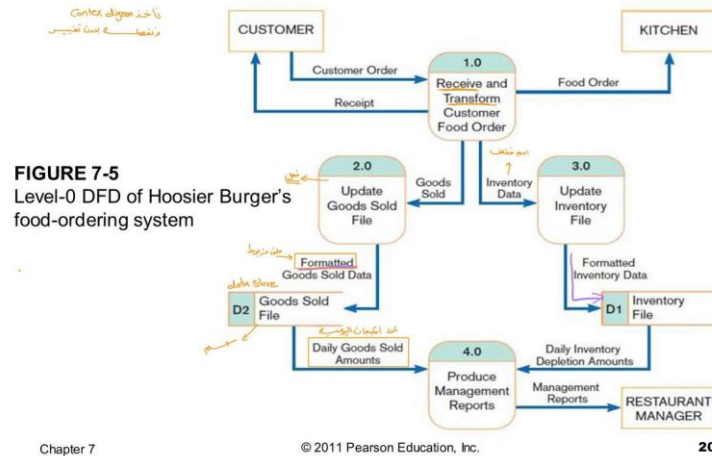


Q6: Draw context from level-0 diagram.

فكرته تعطيك رسمة لزيرو دايقرام وتحوليه إلى كونتيكس

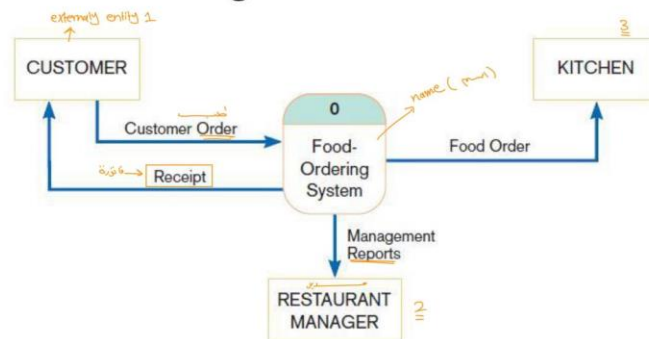
ما جا نفس ذا بس نفس الفكرة

Level-0 Diagram



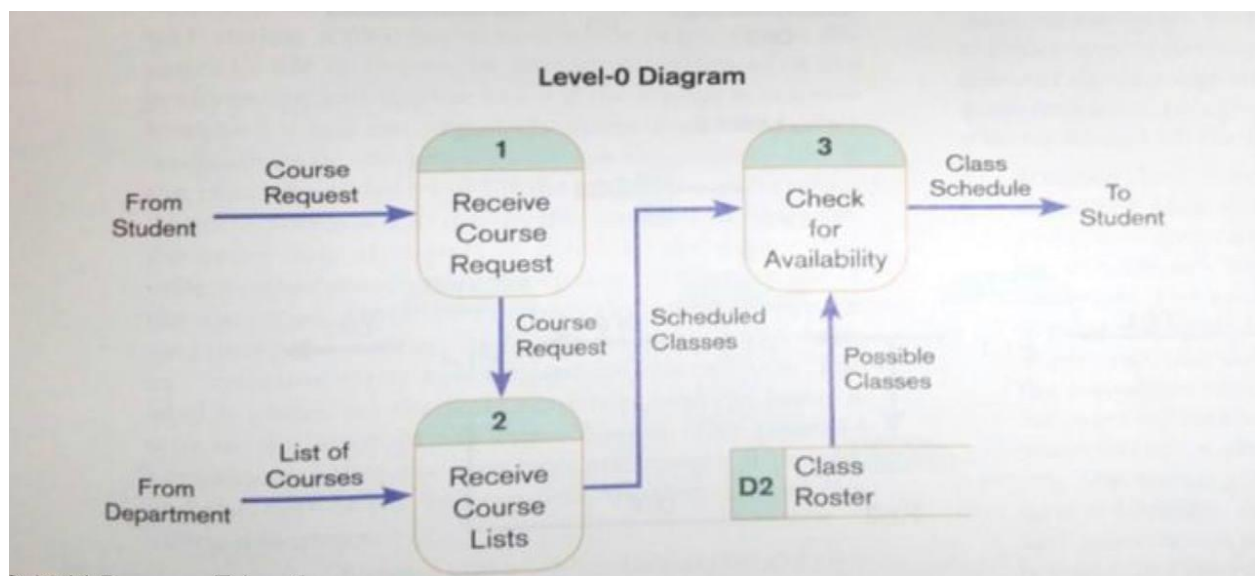
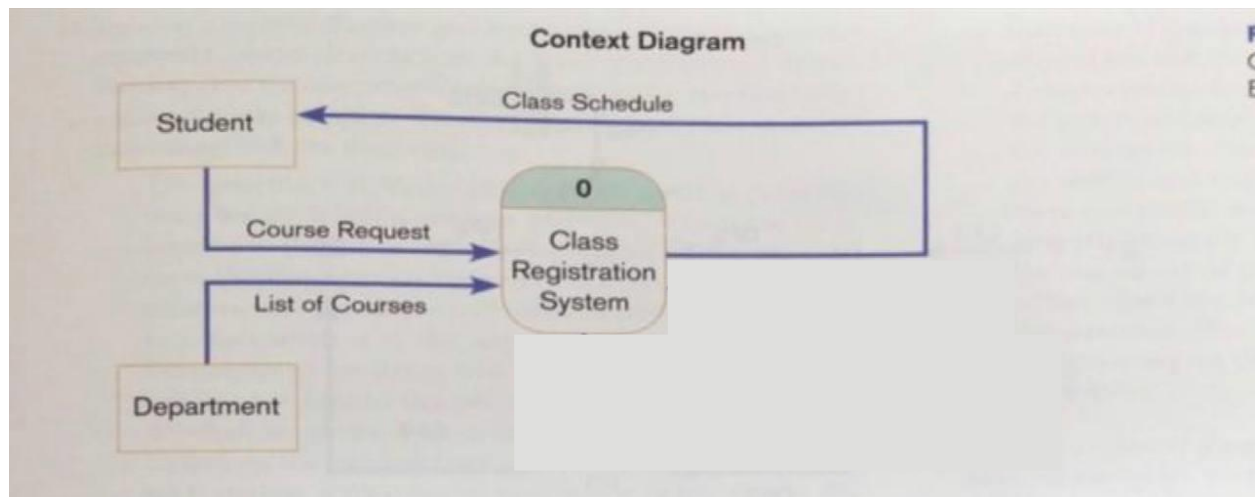
الحل:

Context Diagram



Q7: Draw level-0 diagram from context.

جا نفس ذا بالزبط مو لازم يتسوى نفس الرسمة ارسامي بطريقتك



بالتوفيق ج ~